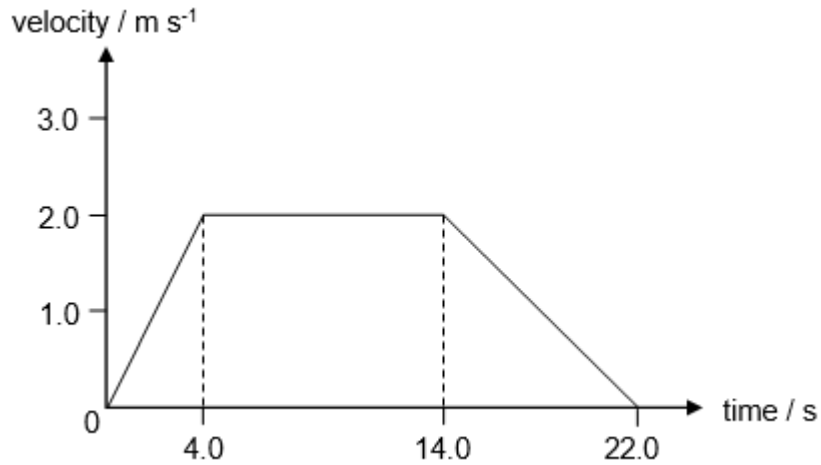


- 8 The diagram shows the velocity-time graph of a 500 kg elevator during its upward motion.



What is the maximum power generated by the engine of the elevator?

- | | | | | | | | |
|----------|--------|----------|--------|----------|--------|----------|---------|
| A | 7810 W | B | 9310 W | C | 9810 W | D | 10300 W |
|----------|--------|----------|--------|----------|--------|----------|---------|

Answer: D

Let T be the upward pull by the elevator cable on the elevator.

Power = Tv

P is max when product Tv is max.

T is max when elevator's acceleration is upwards, i.e. from $t = 0$ s to 4.0 s.

Maximum power generated just before $t = 4.0$ s, **when speed is max**.

$$T - mg = ma$$

$$T = ma + mg = 500 \left(\frac{2.0}{4.0} + 9.81 \right)$$

$$P_{\max} = Tv = 500 \left(\frac{2.0}{4.0} + 9.81 \right) (2.0) = 10310 = 10300 \text{ W}$$

9 Two identical objects P and Q are placed on a flat rough circular disc. Q is at a further